Mengqian Zhang

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Research Interests

My research focuses on the **economic security** of decentralized systems, particularly **blockchain** and **decentralized finance**, using approaches from **mechanism design** and **game theory**.

Professional Experience

Carnegie Mellon University — **Pittsburgh, USA**
Postdoctoral Fellow, Department of Computer Science, hosted by Elaine Shi — Dec. 2025 - Present

Yale University — **New Haven, USA**
Postdoctoral Associate, Department of Computer Science, hosted by Fan Zhang — Feb. 2024 - Nov. 2025

New York University — **New York, USA**
Postdoctoral Researcher, Stern School of Business, hosted by Xi Chen and Elynn Chen — Feb. 2023 - Jan. 2024

Peking University — **Beijing, China**
Visiting Scholar, Center on Frontiers of Computing Studies, advised by Xiaotie Deng — Jul. 2019 - Dec. 2022

Education

Shanghai Jiao Tong University — **Shanghai, China**
Ph.D., Computer Science, advised by Xiaotie Deng — Sep. 2018 - Dec. 2022

- Thesis: Competition and Cooperation in Blockchain Systems: Equilibrium Analysis and Decision Optimization

Ocean University of China — **Qingdao, China**
B.E., Computer Science, Rank: 1/120 — Sep. 2014 - Jun. 2018

Selected Awards/Grants

- **Best DeFi Paper Award** at ACM CCS Workshop on Decentralized Finance and Security (DeFi) 2026
- **Uniswap Foundation TLDR Fellowship** 2024
- **Ethereum Foundation Academic Grants** *(with Prof. Fan Zhang)* 2024
- **IPDPS 2020 Best Paper Nominee** *(4/110 accepted papers)* 2020
- **The First Prize Scholarship** 2017
- **National Scholarship** *(The highest scholarship for Chinese students)* 2015 & 2016

Selected Papers

[1] **Otter: A Provably MEV-Resilient Automated Market Maker via Surplus Redistribution**
Elaine Shi, **Mengqian Zhang**, Hao Chung, and Yuhao Li *(randomized author order)*
In submission, 2026.

[2] **Sybil-Proofness of Generalized Adaptive Clinching and Implications for Blockchain Mechanism Design**
Chengkai Zhang, Yuhao Li, Elaine Shi, **Mengqian Zhang**, Lirong Xia, and Qishen Han *(randomized author order)*
In submission, 2026.

[3] **Tight Characterizations of AMM Mechanism Design under Weak Efficiency**
Chengkai Zhang, Lirong Xia, Elaine Shi, **Mengqian Zhang**, Qishen Han, and Yuhao Li *(randomized author order)*
In submission, 2026.

[4] **A Trilemma in AMM Mechanism Design**
Yuhao Li, Elaine Shi, and **Mengqian Zhang** *(randomized author order)*
Proceedings of the 30th International Conference on Financial Cryptography and Data Security (FC), 2026.

[5] **Boost+: Equitable, Incentive-Compatible Block Building**
Mengqian Zhang[1], Sen Yang[1], Kartik Nayak, and Fan Zhang *([1] co-first author)*
To appear in the Network and Distributed System Security (NDSS) Symposium, 2027.

[6] **Shill Bidding in MEV-Boost Auctions**
Mengqian Zhang, Sen Yang, Thomas Thiery, and Fan Zhang
In submission, 2025.

[7] **RediSwap: MEV Redistribution Mechanism for CFMMs**
Mengqian Zhang, Sen Yang, and Fan Zhang
Proceedings of the 2025 ACM CCS Workshop on Decentralized Finance and Security (DeFi), 2025.

[8] **Maximal Extractable Value in Batch Auctions**
Mengqian Zhang, Yuhao Li, Xinyuan Sun, Elynn Chen, and Xi Chen
Proceedings of the 26th ACM Conference on Economics and Computation (EC), 2025.
Best DeFi Paper Award at ACM CCS Workshop on Decentralized Finance and Security (DeFi), 2026.

[9] **MEV Makes Everyone Happy under Greedy Sequencing Rule**
Yuhao Li[1], **Mengqian Zhang**[1], Jichen Li[1], Elynn Chen, Xi Chen, and Xiaotie Deng *([1] co-first author)*
Proceedings of the 2023 ACM CCS Workshop on Decentralized Finance and Security (DeFi), 2023.

[10] **Insightful Mining Equilibria**
Mengqian Zhang, Yuhao Li, Jichen Li, Chaozhe Kong, and Xiaotie Deng
The 18th Conference on Web and Internet Economics (WINE), 2022.

[11] **Accelerating Transactions Relay in Blockchain Networks via Reputation**
Mengqian Zhang, Yukun Cheng, Xiaotie Deng, Bo Wang, Jan Xie, Yuanyuan Yang, and Jiarui Zhang
IEEE/ACM 29th International Symposium on Quality of Service (IWQoS), 2021.

[12] **CycLedger: A Scalable and Secure Parallel Protocol for Distributed Ledger via Sharding**
Mengqian Zhang, Jichen Li, Zhaohua Chen, Hongyin Chen, and Xiaotie Deng
IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2020.

[13] **Two-Tier Sharing in Electric Vehicle Service Market**
Yukun Cheng, Xiaotie Deng, and **Mengqian Zhang** *(alphabetical author order)*
IEEE Transactions on Cloud Computing, 10(1), pp. 724–735, 2019.

Academic Talks

- **Foundations of Mechanism Design for Decentralized Finance**
  - CMU Theory Lunch 10/21/2026
- **MEV in Batch Auctions**
  - The Twenty-Sixth ACM Conference on Economics and Computation (EC'25), Stanford 7/7/2025
  - The Latest in DeFi Research (TLDR) Conference 2024, Columbia Business School 5/16/2024
  - Yale YACL Reading Group 4/4/2024
- **MEV Makes Everyone Happy under Greedy Sequencing Rule**
  - ACM CCS Workshop on Decentralized Finance and Security, Copenhagen, Denmark 11/30/2023
  - 5th International Conference on Blockchain Economics, Security and Protocols (Tokenomics 2023), Columbia University 10/28/2023
- **MEV on MEV Solutions**
  - Full-Stack Security Lab at Syracuse University (remote) 11/17/2023
  - Yale Computer Science 11/14/2023
- **Insightful Mining Equilibria**
  - 18th Conference on Web and Internet Economics (WINE 2022) (remote) 12/15/2022
  - *Game Theory in Blockchain Session* of International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW 2022) (remote) 8/17/2022
  - 19th Annual Conference of China Society for Industrial and Applied Mathematics (CSIAM 2021), Hefei, China 10/9/2021
- **Accelerating Transactions Relay in Blockchain Networks via Reputation**
  - 17th International Conference on Emerging Technologies for a Smarter World (remote) 11/4/2021
  - 29th International Symposium on Quality of Service (IWQoS 2021) (remote) 6/25/2021
- **CycLedger: A Scalable and Secure Parallel Protocol for Distributed Ledger via Sharding**
  - Singapore University of Technology and Design (remote) 3/26/2021
  - Beijing Normal University 1/10/2020
- **Two-Tier Sharing in Electric Vehicle Service Market**
  - 13th International Frontiers of Algorithmics Workshop (FAW 2019), Sanya, China 4/30/2019

Teaching

Cryptocurrency and Blockchain: Theory to Practice Fall 2026
Guest lecturer for three lectures on Decentralized Finance Carnegie Mellon University
Course instructor: Elaine Shi

Foundations of Blockchains Spring 2026
Guest lecturer for three lectures on Decentralized Finance Carnegie Mellon University
Course instructor: Elaine Shi

Professional Service

- **Program Committee:** FC 2025, EC 2026, AFT 2026, CCS DeFi 2026
- **External Reviewer:**
  - Conference: WINE 2020, ICDCS 2022, WWW 2022, ISAAC 2024, AFT 2024, ITCS 2026, WINE 2026
  - Journal: Operations Research, IEEE JSAC, IEEE TPDS, IEEE TCC
- **Conference Organization:**
  - Co-organizer of the Blockchain Session at IJTCS 2021
  - Organizer of the Student Forum at CSIAM Blockchain Track 2021-2022
  - Co-organizer of the Poster Session at WINE 2020

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